UNDERBOT — LINKED ELECTRICAL SCHEMATIC / REVISION 0.5

Double-click a sheet below to see the actual circuit. Same global net name connects between sheets.

01 — Battery input, isolation and Kelvin shunt

File: 01_Battery.kicad_sch

02 — INA228 current / voltage monitor circuit

File: 02_Monitor.kicad_sch

03 — Fused converter harnesses and power outputs

File: 03_Rails.kicad_sch

04 — Controller and sensor harness / temperature circuit

File: 04_Control.kicad_sch

05 — Actuator power and external safety interfaces — HOLD

File: 05_Actuators.kicad_sch

06 — Single-source external charger input — prototype

File: 06_Charge.kicad_sch

07 — Off-board modules and ESP32 GPIO wiring

File: 07_Modules.kicad_sch

08 — Ten servo signal and fused power branches

File: 08_ServoBranches.kicad_sch

09 — Selected commercial power modules

File: 09_SelectedPower.kicad_sch

Detailed now: fuse / isolator / shunt path, INA228 pins, branch harnesses, sensor bus and NTC divider.

HOLD: physical connectors, converter selection, actuator ratings, safety hardware and OEM base integration.

PROVISIONAL — NOT FOR FABRICATION

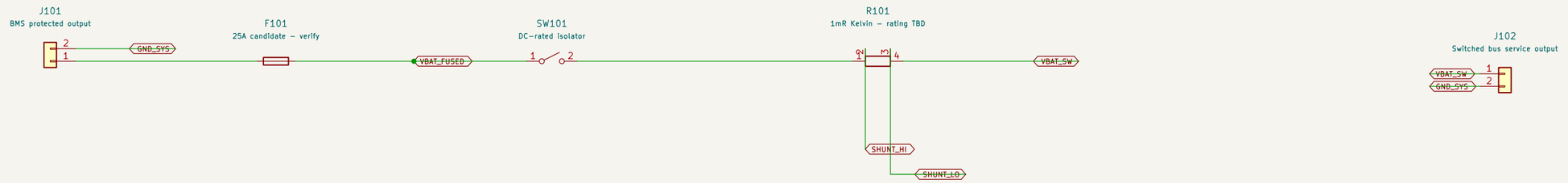
Sheet: /
File: UnderBot.kicad_sch

Title: UnderBot — electrical integration schematic

Size: A3	Date: 2026-10-02	**Rev: 0.5 PROCUREMENT**
KiCad E.D.A. 10.0.6		Id: 1/10

BATTERY -> MAIN FUSE -> SERVICE SWITCH -> SHUNT -> LOAD BUS

Protected 6S pack only. Charge connection branches BEFORE service switch; see sheet 06.



R101 is a FOUR-terminal shunt: pins 1/4 carry current; pins 2/3 are Kelvin sense connections to U101 (sheet 02).

1 milliohm: 25A -> 25mV and 0.625W; 40A -> 1.6W. Select pulse rating, thermal margin and terminal geometry.

OFF-BOARD / NOT RELEASED: cell pack, BMS, pack fuse, isolator and high-current shunt. No DIY cell assembly drawing.

GND_SYS is the BMS protected return. Do not bypass the BMS with USB grounds, charger returns or chassis bonds.

F101 protects the battery lead. Branch fuses are on sheets 03/05/06. No fuse, cable or contactor rating is final.

UNRESOLVED: whole-system reverse-polarity / surge protection and converter inrush. No placeholder is claimed as protection.

SW101 OFF isolates loads, not the charging connector. Disconnect the pack and charger before service.

This shunt measures load discharge only. Charging bypasses it; BMS must supply total pack charge accounting.

J pin numbers are proposed harness assignments, not OEM connector pinouts
PROVISIONAL CIRCUIT / HARNESS DESIGN – NOT FOR FABRICATION

UnderBot

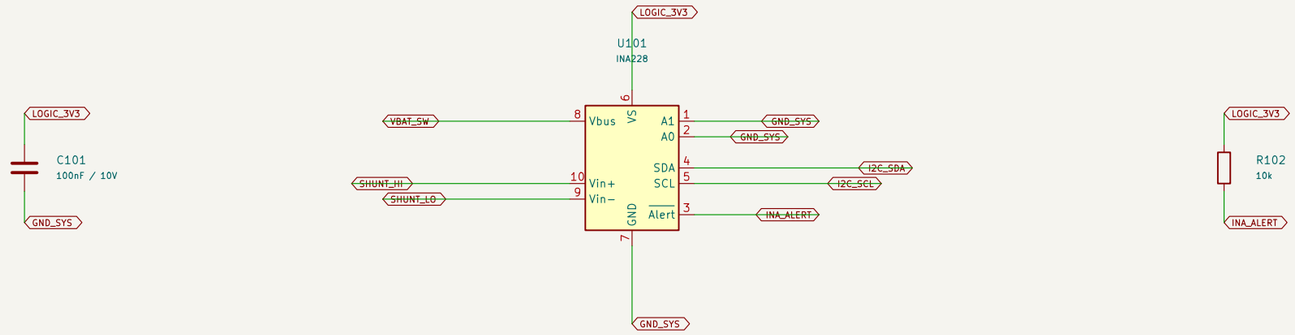
Sheet: /01 – Battery input, isolation and Kelvin shunt/
File: 01_Battery.kicad_sch

Title: 01 – Battery input, isolation and Kelvin shunt

Size: A3	Date: 2026-10-02	**Rev: 0.5 PROCUREMENT**
KiCad E.D.A. 10.0.6		Id: 2/10

REAL IC PIN CONNECTIONS: INA228 / DGS 10-PIN PACKAGE

Prototype may use an equivalent breakout, but check onboard shunt and pin labels before connecting it.



C101 goes directly beside U101 VS/GND. A0=A1=GND selects 0x40; do not put another 0x40 device on the same bus segment.

INA228 VS is 3.3V, NEVER battery voltage. Only VBUS and IN+/IN− connect to the high-voltage measurement path.

Run SHUNT_HI / SHUNT_LO as a matched Kelvin pair from R101 sense terminals. Do not carry load current in those wires.

I2C pull-ups are on sheet 04, subject to existing module pull-ups. ALERT is open-drain and pulled up by R102.

Candidate shunt needs firmware calibration. Verify measurement range, reset behavior and power sequencing before use.

No input filter values are assumed. Evaluate TI input-filter guidance against actual converter / motor noise.

Reference: TI INA228 datasheet, pin-function table and application circuit. Footprint intentionally awaits package verification.

J pin numbers are proposed harness assignments, not OEM connector pinouts
PROVISIONAL CIRCUIT / HARNESS DESIGN — NOT FOR FABRICATION
UnderBot

Sheet: /02 — INA228 current / voltage monitor circuit/
File: 02_Monitor.kicad_sch

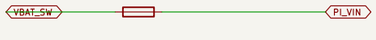
Title: 02 — INA228 current / voltage monitor circuit

Size: A3	Date: 2026-10-02	**Rev: 0.5 PROCUREMENT**
KiCad E.D.A. 10.0.6		Id: 3/10

COMMERCIAL DC/DC MODULES: INPUT AND OUTPUT HARNESS CONNECTIONS

Pairs TO_x and FROM_x go to an off-board converter. They are NOT shorted together on the PCB.

F201
Rating TBD



J211
TO PI converter VIN/GND



J212
FROM PI VOUT/GND

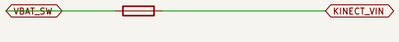


J213
PI load output



Pololu D36V50F12 -> 12V -> 52Pi EP-0225 -> Pi USB-C; see sheet 09

F202
Rating TBD



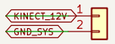
J221
TO KINECT converter VIN/GND



J222
FROM KINECT VOUT/GND

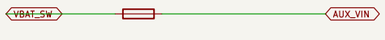


J223
KINECT load output



Pololu D36V50F12 -> Xbox 360 Kinect v1 power adapter (verify adapter label)

F203
Rating TBD



J231
TO AUX converter VIN/GND



J232
FROM AUX VOUT/GND



J233
AUX load output



Pololu D36V28F5: 5V auxiliaries; design allocation 2A continuous

J241
TO 3.3V regulator VIN/GND



J242
FROM 3.3V regulator VOUT/GND



#FLG01
PWR_FLAG



#FLG02
PWR_FLAG



All J numbers define OUR harness ends. Verify converter terminal labels, polarity and connector current limits.

J213 is 12V into EP-0225 DC input, NEVER into Pi USB. Use its supplied USB-C output cable. Raw 6S exceeds its 24V limit.

J pin numbers are proposed harness assignments, not OEM connector pinouts
PROVISIONAL CIRCUIT / HARNESS DESIGN — NOT FOR FABRICATION
UnderBot

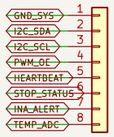
Sheet: /03 — Fused converter harnesses and power outputs/
File: 03_Rails.kicad_sch

Title: 03 — Fused converter harnesses and power outputs

Size: A3	Date: 2026-10-02	**Rev: 0.5 PROCUREMENT**
KiCad E.D.A. 10.0.6		Id: 4/10

ESP32 IS USB-POWERED FROM THE PI. ALL SIGNAL LEVELS HERE ARE 3.3V.

J301
ESP32 signal harness - see sheet 07



J302
Adafruit 2717 mux - 0x77



J303
Body: Adafruit 4554 / channel 0



J305
Mux channels 0 / 1 outputs

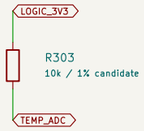


J304
Head: Adafruit 4554 / channel 1

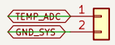








J306
External 10k NTC





J306 NTC + R303 form an ADC divider; C301 filters noise. Confirm thermistor curve and ESP32 ADC range/calibration.

IMUs: ICM-20948 breakouts, polled I2C. J303/J304 pins 1/2/3/4 -> VIN/GND/SDA/SCL. Software fusion required.

The mux is OFF-BOARD. Its channel ports are J305; it is not replaced by shorting the SDA/SCL lines together.

Mux A0/A1/A2 -> 3.3V: 0x77 avoids PCA9685 ALLCALL 0x70. Select ONE channel at a time. Not Qwiic pin numbering.

Set PCA9685 A0 jumper CLOSED: 0x41. A mux channel alone does NOT avoid upstream INA228 address 0x40.

J pin numbers are proposed harness assignments, not OEM connector pinouts
PROVISIONAL CIRCUIT / HARNESS DESIGN - NOT FOR FABRICATION

UnderBot

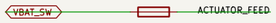
Sheet: /04 - Controller and sensor harness / temperature circuit/
File: 04_Control.kicad_sch

Title: 04 - Controller and sensor harness / temperature circuit

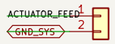
Size: A3	Date: 2026-10-02	**Rev: 0.5 PROCUREMENT**
KiCad E.D.A. 10.0.6		Id: 5/10

ACTUATOR / SAFETY INTERFACES ONLY: HARDWARE STILL TO BE SELECTED

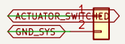
F401
Actuator branch — TBD



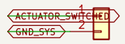
J401
TO external NO contactor LINE



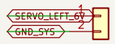
J402
FROM contactor LOAD



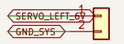
J403
TO two fused BEC inputs (sheet 09)



J404
FROM LEFT Hobbywing BEC 6V



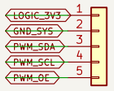
J405
TO LEFT arm + camera pan



J406
Safety-controller logic interface



J407
PCA9685 logic — A0 closed / 0x41



J408
FROM dedicated mux channel 2





Safety controller, two NC E-stop channels, reset, watchdog hardware, coil driver and suppression are NOT DESIGNED here.

J401/J402 are external contactor power terminals, not a PCB jumper. Do not bypass the missing contactor or safety circuit.

PWM_OE HIGH should disable output; resolve breakout pull-downs before fitting R401. Signal disable is not an E-stop.

Two Hobbywing 30606000 BECs at 6.0V. RIGHT rail and J915 on sheet 09. Outputs must NEVER be paralleled.

J pin numbers are proposed harness assignments, not OEM connector pinouts
PROVISIONAL CIRCUIT / HARNESS DESIGN — NOT FOR FABRICATION

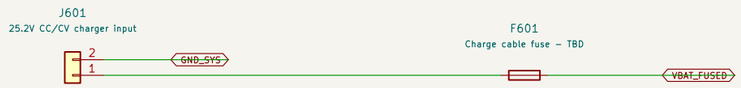
UnderBot

Sheet: /05 — Actuator power and external safety interfaces — HOLD/
File: 05_Actuators.kicad_sch

Title: 05 — Actuator power and external safety interfaces — HOLD

Size: A3	Date: 2026-10-02	**Rev: 0.5 PROCUREMENT**
KiCad E.D.A. 10.0.6		Id: 6/10

FIRST REVISION: ONE APPROPRIATE 6S CHARGER CONNECTED AT A TIME



Uses the common-port BMS topology: charge positive -> F601 -> F101 -> pack positive; return -> protected BMS return.

If the selected BMS has a separate C- port, this sheet MUST change. Never wire charger return directly to raw cell negative.

Selected charger family: Mascot 2440 LI, 6-cell / 25.2V version. Pack supplier to provide matched connector and temperature option.

Prototype charging procedure: service switch OFF; appropriate pack/BMS fitted; one source; no energized motion.

NOT IN THIS REVISION: unattended dock contacts/interlock, parallel solar + mains sources, charge-source OR-ing or supercapacitors.

Station PV / LiFePO4 / buck-boost charger remains an EXTERNAL subsystem in the architecture drawing. No station PCB is implied.

The original T12 dock/battery is independent. Neither T12 battery terminals nor its 20V charging input accepts 25.2V directly.

Mechanical connector selection, touch protection and fuse coordination are still required before connecting real batteries.

J pin numbers are proposed harness assignments, not OEM connector pinouts
PROVISIONAL CIRCUIT / HARNESS DESIGN – NOT FOR FABRICATION
UnderBot

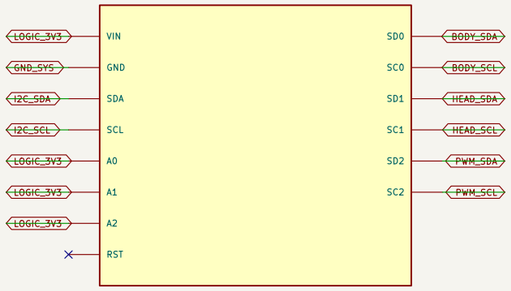
Sheet: /06 – Single-source external charger input – prototype/
File: 06_Charge.kicad_sch

Title: 06 – Single-source external charger input – prototype

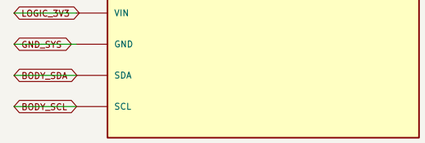
Size: A3	Date: 2026-10-02	**Rev: 0.5 PROCUREMENT**
KiCad E.D.A. 10.0.6		Id: 7/10

MODULE INTERFACES: CONNECT BY BOARD LABEL, NOT BARE-IC PIN NUMBER

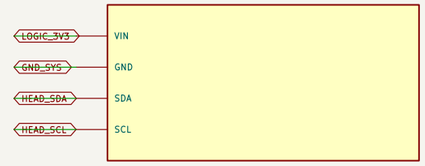
A701
Adafruit 2717 / 0x77



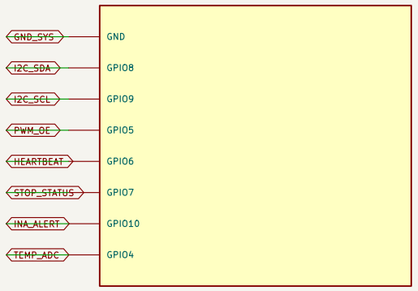
A702
Body IMU / Adafruit 4554



A703
Camera IMU / Adafruit 4554



A704
ESP32-S3-DevKitC-1 / used pins



ESP32 J1 header mapping (official DevKitC-1):
GND=22; GPIO8=12; GPIO9=15; GPIO5=5;
GPIO6=6; GPIO7=7; GPIO10=16; GPIO4=4.
Pi USB -> board USB-to-UART port supplies power/data.
Do not connect its 3V3 output to LOGIC_3V3.

Mux RST left at onboard pull-up; channels 3-7 unused.
IMUs: polled I2C; unused breakout pads not shown.
Select ONE mux channel at a time; software fusion required.
No bare sensor IC or custom mux PCB is implied.

Electrical connection labels span sheets 04/05/07/08. Module symbols expose only the used interfaces.

HOLD: power-off backfeed, ADC attenuation, cable timing, hardware stop and OE startup state must be validated.

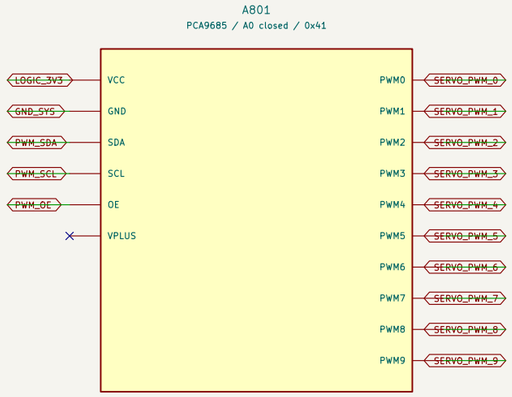
J pin numbers are proposed harness assignments, not OEM connector pinouts
PROVISIONAL CIRCUIT / HARNESS DESIGN — NOT FOR FABRICATION
UnderBot

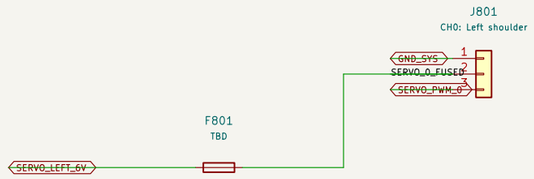
Sheet: /07 — Off-board modules and ESP32 GPIO wiring/
File: 07_Modules.kicad_sch

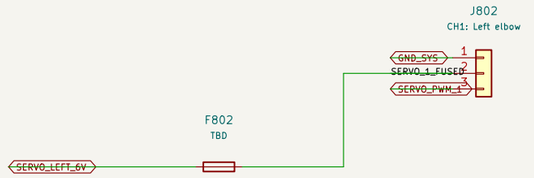
Title: 07 — Off-board modules and ESP32 GPIO wiring

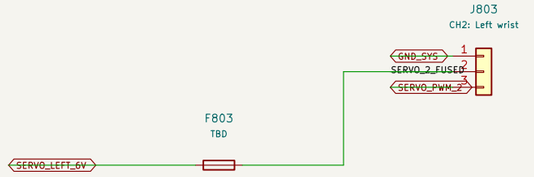
Size: A3	Date: 2026-10-02	**Rev: 0.5 PROCUREMENT**
KiCad E.D.A. 10.0.6		Id: 8/10

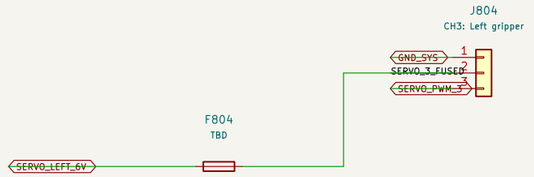
PROVISIONAL: 2 x (3 ARM JOINTS + GRIPPER) + CAMERA PAN/TILT

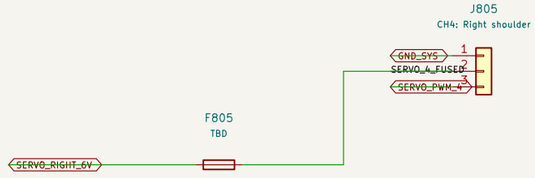


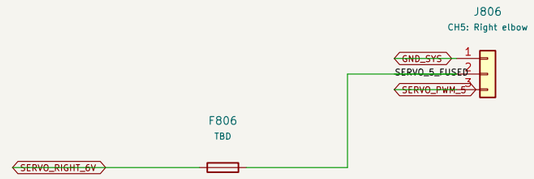


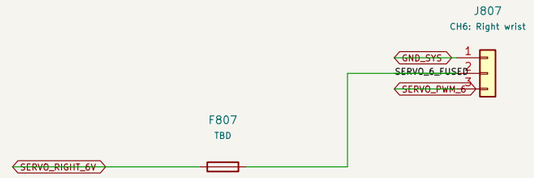


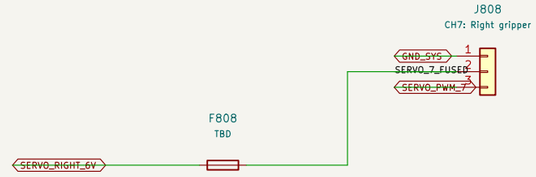


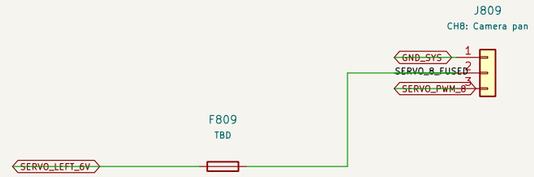


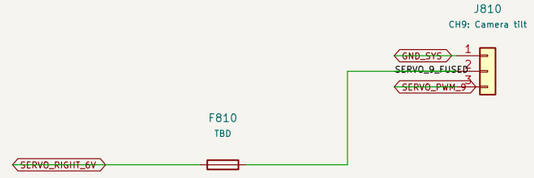












A801 V+ stays unconnected.
Only PWM signals come from this board.
High-current power uses separate
fused distribution and return wiring.

J801–J810: OUR harness convention
1=GND, 2=fused power, 3=signal.
Confirm actual servo plug orientation.
CH10–15 unused; not represented.

Selected: FT5335M shoulders CH0/4; FT5121M all other channels. Target: 190mm reach / 250g payload, subject to load test.

LEFT: CH0/1/2/3/8. RIGHT: CH4/5/6/7/9. Two independent regulated 6.0V positives; shared reference ground.

Power removal may let arms fall. Complete hardware stop, restraints/braking, startup bias and pulse calibration before operation.

J pin numbers are proposed harness assignments, not OEM connector pinouts
PROVISIONAL CIRCUIT / HARNESS DESIGN — NOT FOR FABRICATION

UnderBot

Sheet: /08 – Ten servo signal and fused power branches/
File: 08_ServoBranches.kicad_sch

Title: 08 – Ten servo signal and fused power branches

Size: A3	Date: 2026-10-02	**Rev: 0.5 PROCUREMENT**
KiCad E.D.A. 10.0.6		Id: 9/10

SELECTED POWER MODULES / NO PARALLEL REGULATOR OUTPUTS

A901
D36V50F12 / Pi preregulator



A902
52Pi EP-0225 / Pi5 PD source



USB_C_OUT denotes the COMPLETE supplied cable,
including PD signalling. Never make a two-wire USB adapter.
Use DC input ONLY; 12V preregulation is mandatory.

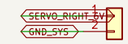
A903
D36V50F12 / Kinect 12V



A908
Raspberry Pi 5 / power cable port



J915
RIGHT distribution



A904
D36V28F5 / auxiliary 5V



A905
D24V5F3 / logic 3.3V



F901
Input fuse TBD



F902
Input fuse TBD



A906
Hobbywing 30606000 / SET 6.0V



A907
Hobbywing 30606000 / SET 6.0V



BEC backup leads: individually insulated, no battery fitted. A backup battery would bypass the main actuator disconnect.

BEC multiple output connectors distribute current; never send all current through one servo plug. Use vendor-rated harnesses.

J pin numbers are proposed harness assignments, not OEM connector pinouts
PROVISIONAL CIRCUIT / HARNESS DESIGN — NOT FOR FABRICATION

UnderBot

Sheet: /09 — Selected commercial power modules/
File: 09_SelectedPower.kicad_sch

Title: 09 — Selected commercial power modules

Size: A3	Date: 2026-10-02	**Rev: 0.5 PROCUREMENT**
KiCad E.D.A. 10.0.6		Id: 10/10